

CC-TECHNO 32

Stage 2 Validation Report

EnerGreen: Green Energy Consumption Monitoring System

Team Name: GreenTechs

Team Member	Role (Hustler, Hacker, Hipster)
Alesna, Christine Anne	Hustler
Amaro, Kobe Bryan	Hacker
Gorgonio, Bryl Darel	Hacker
Postrero, Angela	Hipster

Summary

The Stage 2 validation for EnerGreen focused on testing the viability of our proposed IoT-based energy monitoring solution and solar integration features. With 111 respondents evaluating high-fidelity mockups of the EnerGreen interface, findings revealed strong interest in real-time energy tracking, with 88.2% of respondents believing it would help reduce electricity costs. Notably, 96.4% considered appliance-specific consumption breakdown either "very important" or "somewhat important," and 91.8% showed interest in predicting monthly bills based on usage patterns. Furthermore, 94.6% expressed likelihood to consider solar panels when provided with clear savings estimates, and 97.3% indicated willingness to invest in an IoT-based monitoring device. These findings confirm strong market demand for EnerGreen's proposed solutions and validate our decision to proceed with development of both energy monitoring and solar simulation features.

Validation Objectives

- **Validate the potential effectiveness of EnerGreen's proposed IoT-based energy monitoring system in helping users track and reduce electricity consumption.**

We aimed to assess whether users believe real-time energy tracking would help reduce their electricity costs and if they value features such as appliance-specific consumption breakdowns. This objective was crucial for confirming the core value proposition of our monitoring system.

- **Assess the demand for EnerGreen's solar energy simulation tool among homeowners considering solar panel adoption.**

We sought to understand if providing clear savings estimates and accurate simulations would increase homeowners' likelihood of adopting solar energy solutions. This objective directly informs the development priority of our solar integration features.

- **Determine if users find EnerGreen's AI-powered analytics and carbon footprint tracking valuable for energy optimization.**

We wanted to evaluate whether supplementary features like carbon footprint tracking and predictive bill estimation would influence users' energy consumption habits. This objective helps prioritize feature development beyond core monitoring capabilities.

Methodology

The validation was conducted through an **online survey using Google Forms**, with **111 respondents** from various income levels (ranging from low- to upper-income homeowners). Participants were given the option to either explore the **high-fidelity mockup via Figma** or **watch a pre-recorded demonstration** of the EnerGreen interface, which showcased key features such as real-time energy tracking, appliance consumption breakdown, ai-driven

energy-saving insights, solar savings simulation, and educational resources on energy conservation and efficiency. The questionnaire included 11 questions focusing on current energy tracking methods, interest in real-time monitoring, perceived value of specific features, likelihood of adopting solar energy, and price sensitivity. Quantitative (multiple-choice) questions were used to gather comprehensive insights.

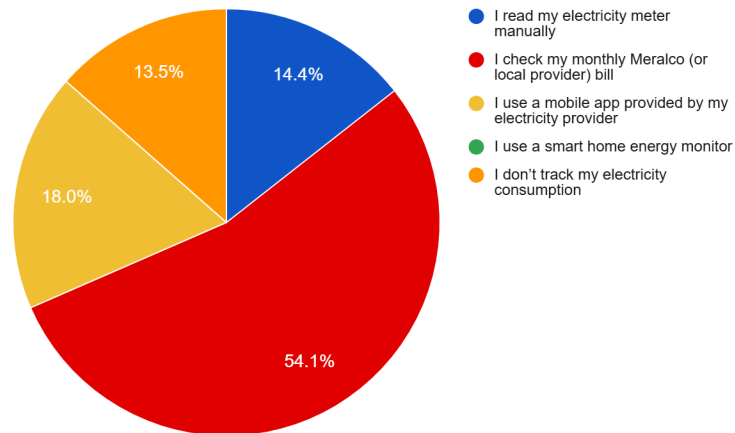
Stage 2 - Validation Board

Startup Name:	Energreen
Customer Segment	Low to Upper Income Homeowners
Problem(s)	Lack of real-time monitoring, unpredictable bills, and no clear solar savings insights.
Solution	EnerGreen provides an IoT-powered dashboard with real-time tracking, AI-driven cost insights, and solar savings simulations.
Experiment	Online survey with a high-fidelity mockup of EnerGreen's interface.
Success Metric	At least 70% of survey respondents are interested in using EnerGreen's product after seeing the prototype.
Result	The survey shows that 92.7% of respondents expressed interest in using EnerGreen's services, with 46.8% being "very interested" and 46% wanting to learn more. Only 7.2% were unsure or not interested. Additionally, 94.6% indicated they would likely consider solar energy adoption if provided with accurate savings simulations, and 88.2% believed real-time energy tracking would help reduce their electricity costs.
Action	PERSEVERE: Our validation results strongly exceed our success metric of 70% interest. We will proceed with developing the EnerGreen IoT monitoring system with integrated solar simulation features, focusing on the price point of ₱1,000-₱3,000 which appeals to the largest segment (46%) of potential customers.

All Insights

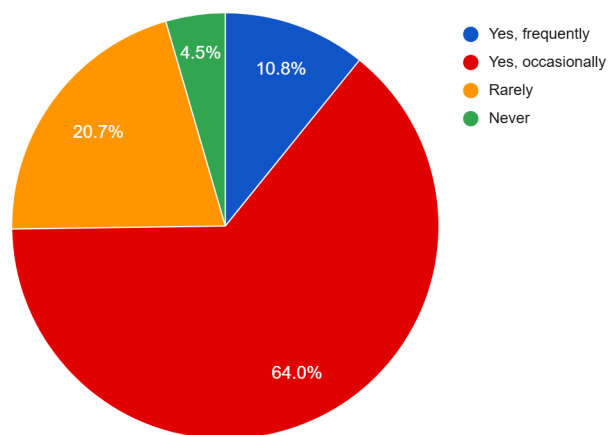
Based on the survey data from 111 respondents, we've identified several key insights:

Current Tracking Methods



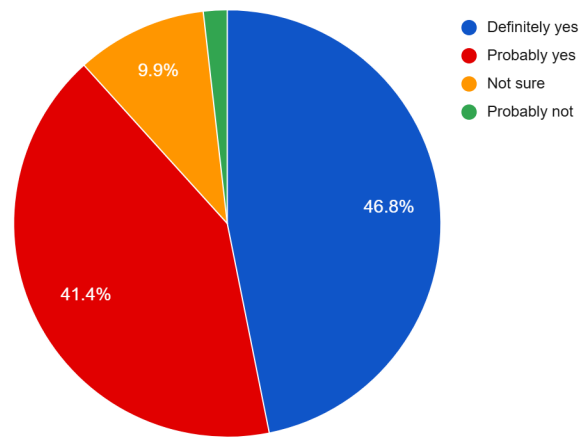
- Only 14.4% read their electricity meter manually
- 54.1% rely solely on monthly utility bills
- 18% use mobile apps provided by electricity providers
- 13.5% don't track electricity consumption at all
- 0% use smart home energy monitors, indicating a clear market gap

Experience with Bill Spikes



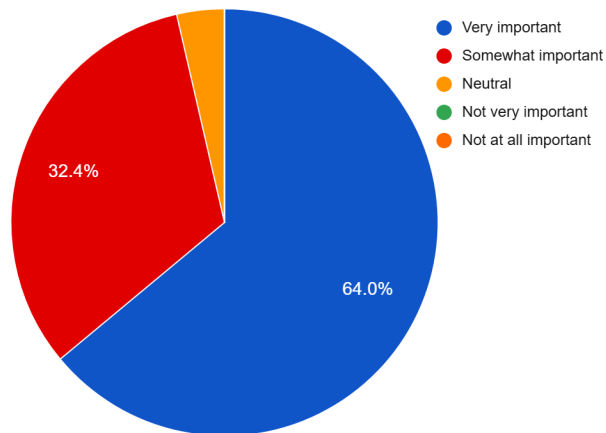
- 74.8% experience unexpected bill spikes (10.8% frequently, 64% occasionally)
- Only 4.5% never experience unexpected changes in bills

Perceived Value of Real-time Tracking



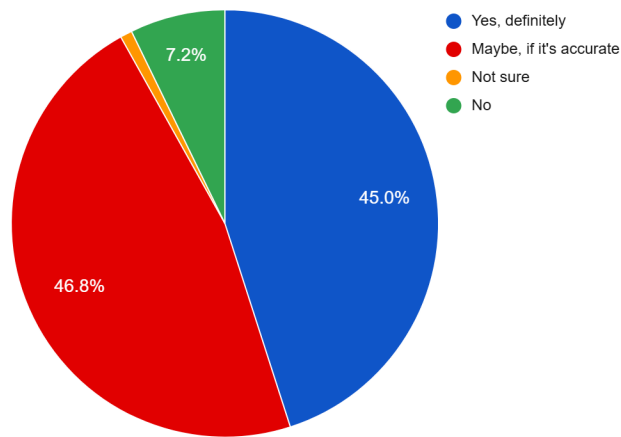
- 88.2% believe real-time tracking would help reduce costs (46.8% "definitely", 41.4% "probably")
- Only 1.8% think it would not help

Importance of Appliance-specific Breakdown



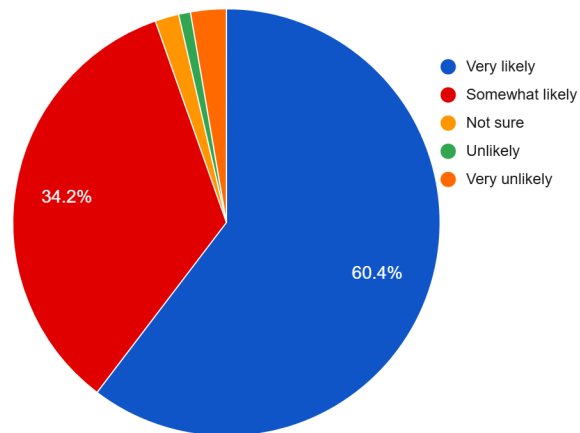
- 64% consider appliance-specific consumption breakdown "very important"
- 32.4% consider it "somewhat important"
- 3.6% are neutral
- 0% consider it unimportant

Interest in Bill Prediction



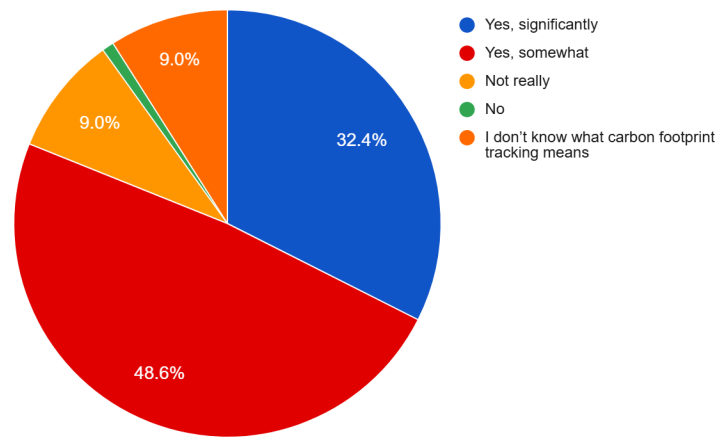
- 45% would "definitely" use a bill prediction tool
- 46.8% would use it if accurate
- Only 8.2% are unsure (7.2%) or uninterested (1%)

Solar Panel Consideration



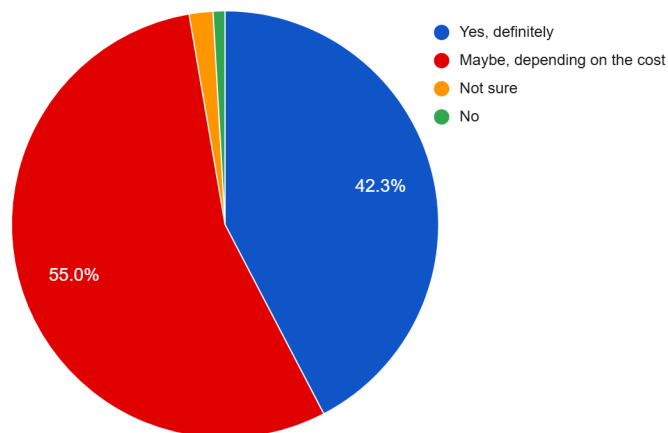
- 60.4% are "very likely" to consider solar panels with clear savings estimates
- 34.2% are "somewhat likely"
- Only 5.4% are unsure (1.8%), unlikely (0.9%) or very unlikely (2.7%)

Carbon Footprint Tracking Impact



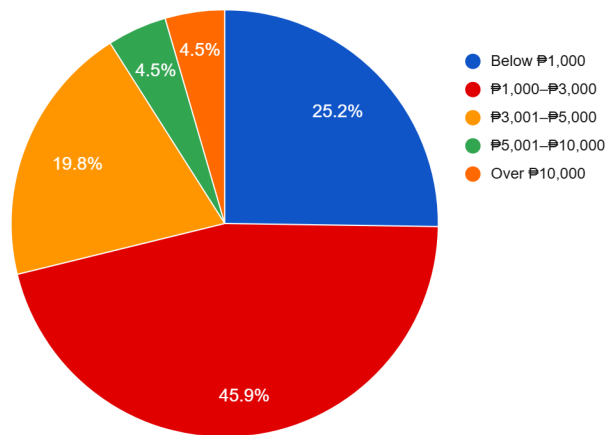
- 32.4% say carbon tracking would significantly influence habits
- 48.6% say it would somewhat influence habits
- 9% say it would not really influence habits
- 9% don't understand carbon footprint tracking

IoT Device Investment Willingness



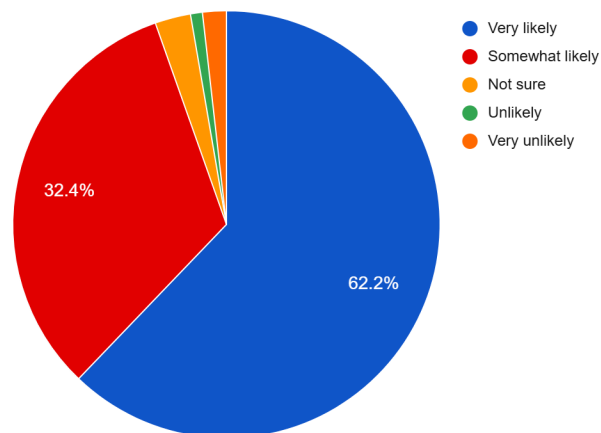
- 42.3% would "definitely" invest in an IoT monitoring device
- 55% might invest depending on cost
- Only 2.7% are unsure (1.8%) or unwilling (0.9%)

Price Sensitivity



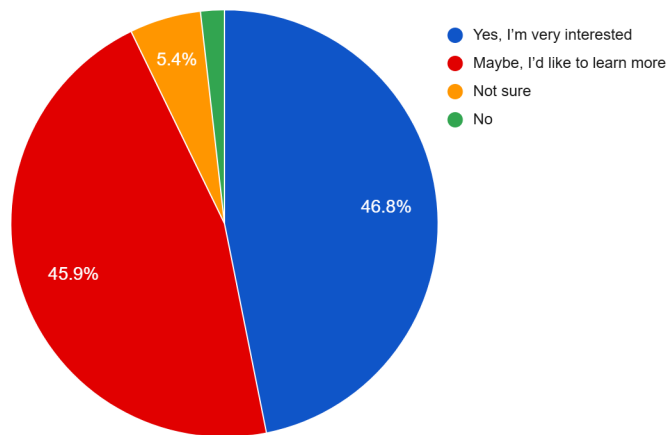
- 25.2% prefer pricing below ₱1,000
- 46% consider ₱1,000-₱3,000 reasonable
- 19.8% accept ₱3,001-₱5,000
- 9% are open to pricing above ₱5,000

Solar Adoption Likelihood



- 62.2% are "very likely" to switch to solar with accurate simulations
- 32.4% are "somewhat likely"
- Only 5.4% are unsure (2.7%), unlikely (0.9%) or very unlikely (1.8%)

EnerGreen Service Interest



- 46.8% are "very interested" in signing up
- 45.9% "might be interested" with more information
- Only 7.2% are unsure (5.4%) or uninterested (1.8%)

Opportunities

Based on our validation results, we've identified several promising opportunities that align with both user demand and market gaps:

1. Smart Home Energy Monitor with AI Insights

With 0% of respondents currently using smart home energy monitors despite high interest (92.8%), there is a significant opportunity to enter the market with an accessible IoT monitoring solution that provides real-time insights into energy consumption. The addition of AI-driven insights differentiates our offering from standard utility tracking methods and existing solutions.

2. Integrated Solar Decision Platform

The overwhelmingly positive response (94.6%) to solar adoption with clear savings estimates reveals a strong opportunity for an integrated platform that not only monitors current consumption but also simulates potential solar savings, serving as a decision-support tool for homeowners considering renewable energy.

3. "Energy Bill Anxiety" Solution

With 74.8% experiencing unexpected bill spikes, there's a clear opportunity to position EnerGreen as a solution for "energy bill anxiety" through predictive analytics and real-time monitoring that helps users avoid surprises and manage costs proactively.

4. Tiered Service Model

The varied price sensitivity (25.2% preferring under ₱1,000, 46% finding ₱1,000-₱3,000 reasonable, and 28.8% willing to pay more) suggests an opportunity for a tiered service model that caters to different budget constraints while providing value at each price point.

5. Eco-conscious Consumer Education

The interest in carbon footprint tracking (81% indicating it would influence habits) reveals an opportunity to incorporate educational content and gamification elements that appeal to environmentally-conscious consumers while driving engagement with the platform.

Recommendations

Based on the insights gathered, we recommend the following actions to move forward with EnerGreen:

1. Develop a Functional MVP

Create a minimum viable product focusing on the most valued features:

- Real-time electricity consumption tracking (daily to yearly views)
- Smart appliance-level consumption breakdown and monitoring
- AI-driven insights and monthly bill prediction
- Comprehensive solar insights and savings simulation
- Educational resources on energy conservation

This MVP should be priced within the ₱1,000-₱3,000 range to capture the largest segment (46%) of potential customers while providing sufficient value to justify the investment.

2. Prioritize Feature Development Based on User Value

Our development roadmap should prioritize features according to validated user demand:

1. Appliance consumption breakdown (rated "very important" by 64%)

2. Monthly bill prediction tools (92% interest)
3. Solar savings simulation (94.6% likelihood to consider solar with this feature)

This approach ensures we're building what users value most while managing development resources efficiently.

3. Implement a Two-Phase Go-to-Market Strategy

Phase 1: Energy Monitoring Focus

- Launch the core IoT monitoring system targeting the 74.8% who experience unexpected bill spikes
- Position as an affordable solution to "energy bill anxiety"
- Emphasize real-time tracking and bill prediction capabilities

Phase 2: Solar Integration Expansion

- Once the monitoring system gains traction, introduce enhanced solar simulation features
- Target the 94.6% of users who expressed interest in solar adoption with clear savings estimates
- Partner with solar panel providers to create an ecosystem of energy solutions

4. Create Educational Content and Onboarding Materials

Since 54.1% of respondents rely solely on monthly bills and 13.5% don't track consumption at all, we should:

- Develop clear, jargon-free educational materials about energy consumption
- Create an intuitive onboarding process that helps users understand their energy usage
- Provide regular tips and insights based on actual consumption patterns

5. Conduct Limited-Scale Pilot Testing

Before full-scale production:

- Deploy functional prototypes with 15-20 households representing different segments
- Gather real-world usage data and feedback over a 2-month period
- Refine the product based on actual user experiences and technical performance

By following these recommendations, we can build a market-ready product that addresses validated user needs while maintaining a realistic development and business approach. The strong validation results support moving forward with EnerGreen as a sustainable business opportunity with significant potential for social impact through energy efficiency and renewable energy adoption.

Relevant Links

<https://drive.google.com/drive/folders/1OJhx-na61bCvKqU7ayCNi8IlmEIDaGFB?usp=sharing>

Figma Prototype:

<https://www.figma.com/proto/l8Ss5z4NXkWNUELbm5Khxr/HCI---Green-Power-System?node-id=1086-10677&t=euGAz5FctQ38B8AL-1&scaling=scale-down&content-scaling=fixed&page-id=672%3A3965&starting-point-node-id=1086%3A10677>